Quantum Gravitational Dynamics: Post-Newtonian Programme v3.0

Resolution of the g_{rr} Problem, Ansatz C Extended to 8PN,
$\zeta(3)$ Local/Nonlocal Distinction, and Corrected Predictions

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March 2026 | DOI: 10.5281/zenodo.18827993

Abstract

We present updated results in the Quantum Gravitational Dynamics (QGD) post-Newtonian programme. Three open problems from v2 are resolved or significantly advanced: (1) The $g_{rr} = 1$ issue in the master metric is resolved: a radial component $\sigma_r = \sigma_t/\sqrt{f}$ with $\varepsilon_r = +1$ gives $g_{rr} = 1/f$ exactly; the enforcement $g_{rr} = -1/g_{tt}$ (FIX 3) is the exact coordinate relation valid for all diagonal spherically-symmetric metrics, not an *ad hoc* choice. (2) Ansatz C is verified and extended to 8PN ($n = 9$). The key results of v2 are confirmed unchanged. (3) The $\zeta(3)$ question is clarified from the GR literature: $\zeta(3)$ enters at 6PN in the *nonlocal* tail sector (scattering angle, Delaunay Hamiltonian) [4], but is absent from the *local* $A(u;\nu)$ potential at 6PN. This means our prediction for the 6PN π^6 coefficient remains valid. The corrected constraint $c_{60}^{\text{tot}} \sim O(10\text{–}50)$ replaces the unverified $c_{60}^{\text{tot}} \approx 4$; the rational chain ($c_{70}^{\text{rat}}$, etc.) is withdrawn.

Critical prediction (unchanged from v2): $c_{60}^{\pi^6}/\nu = -5175625/335872$.

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1 Introduction

This paper is the third in a series developing the QGD post-Newtonian programme [1]. We address three specific criticisms raised in v2.

The GR frontier as of March 2026: the local 6PN dynamics is known except for four ν^3 coefficients [3]; the 6.5PN tail sector was completed in July 2025 [2]; local 7PN and above remain open.

What this paper settles.

- g_{rr} issue: resolved. FIX 3 is exact, not *ad hoc*.
- $\zeta(3)$: nonlocal at 6PN. Does not break our 6PN prediction.
- Ansatz C extended to 8PN.
- c_{60}^{tot} range corrected; rational chain withdrawn.

2 The g_{rr} Problem Resolved

2.1 The issue

The QGD master metric with $\sigma_\mu = (\sigma_t, 0, 0, 0)$ gives $g_{tt} = -(1-\sigma_t^2)$ and $g_{rr} = 1$ (isotropic gauge). The Schwarzschild metric in standard coordinates has $g_{rr} = 1/(1 - r_s/r) \neq 1$.

2.2 Resolution via additional σ_r component

Theorem 2.1 ($g_{rr} = 1/f$ from σ_r)**.** *Let $\sigma_t = \sqrt{r_s/r}$, $f = 1 - r_s/r$, and define the radial component $\sigma_r = \sigma_t/\sqrt{f}$ with signature $\varepsilon_r = +1$. Then the QGD master metric gives*

$$g_{rr} = 1 + \sigma_r^2 = 1 + \frac{\sigma_t^2}{f} = 1 + \frac{1-f}{f} = \frac{1}{f}. \tag{1}$$

Verified numerically: $g_{rr} = 1/(1 - r_s/r)$ to machine precision at all $r > r_s$.

Remark 2.2. $\sigma_r = \sigma_t/\sqrt{f}$ diverges at $r = r_s$, correctly reflecting the coordinate singularity of Schwarzschild curvature coordinates at the horizon. In isotropic coordinates, $\sigma_r = 0$ and $g_{rr} = 1$ is the natural form.

2.3 FIX 3 is the exact Birkhoff relation

In any diagonal spherically-symmetric metric with $g_{tr} = 0$,

$$g_{tt} \cdot g_{rr} = -1 \qquad \text{(Schwarzschild-coordinate identity)} \tag{2}$$

follows from the Birkhoff theorem and the determinant condition $\det(g_{\mu\nu}) = -r^4 \sin^2\theta$. Equation (2) is what FIX 3 enforces. It is exact for all diagonal spherically-symmetric solutions

(Schwarzschild, Reissner–Nordström, Schwarzschild–de Sitter, etc.) in curvature gauge. The isotropic gauge ($g_{rr} = 1$) is equally valid; the two are related by a coordinate transformation.

Table 1: g_{rr} across 7 coordinate representations. All exact in QGD.

Solution	σ_r required	g_{rr} result
Schwarzschild (isotropic)	0	1 [QGD default]
Schwarzschild (curvature)	$\sigma_t/\sqrt{f}$	$1/f$
Kerr–Schild (ingoing)	σ_t (null vector)	1, with $g_{tr} = -r_s/r$
Reissner–Nordström	$(\sigma_M - \sigma_Q)/\sqrt{f_{\mathrm{RN}}}$	$1/f_{\mathrm{RN}}$
de Sitter	Hr/c (radial, $\varepsilon = +1$)	$1/(1 - H^2r^2/c^2)$
Anti-de Sitter	$\lvert H\rvert r/c$ (radial, $\varepsilon = -1$)	$1/(1 + H^2r^2/c^2)$
Kerr (Boyer–Lindquist)	analysis in progress	Σ/Δ [partial]

3 Ansatz C Extended to 8PN

3.1 The correct structure (from v2)

The complete transcendental diagonal of $A(u;\nu)$ at nPN is

$$\boxed{\left[A_{\mathrm{trans}}\right]_n = \frac{c_{n0}}{\nu}\,\pi^{2(n-3)} \sum_{k=0}^{n-4} \beta^{2k}\,\nu^{k+1}, \qquad \frac{c_{n0}}{\nu} = \left(-\frac{41}{32}\right)\left(-\frac{2275}{656}\right)^{n-4}.} \tag{3}$$

Key identity: $32 \times 656 = 512 \times 41 = 20992$.

3.2 Extension to $n = 9$ (8PN)

All $n = 4, \ldots, 9$ have been verified: the diagonal coefficient matches the Padé ladder, and the ν-power equals ν^{n-3} (loop-counting theorem).

Table 2: Padé diagonal $c_{n0}/\nu = (-41/32)(-2275/656)^{n-4}$ and ν-tower through $n = 9$ (8PN)

n	PN	c_{n0}/ν (exact)	$c_{n0}\pi^{2(n-3)}/\nu$	GR status
4	3PN	$-41/32$	-12.645	DJS 2001 ✓
5	4PN	$+2275/512$	$+432.82$	BD/DJS 2015 ✓
6	5PN	$-5175625/335872$	-14814.54	**PREDICTED**
7	6PN	$+11774546875/220332032$	$+507067.23$	**PREDICTED**
8	7PN	$-26787094140625/144537812992$	-17355728.66	**PREDICTED**
9	8PN	$+60940639169921875/94816805322752$	$+594046113.40$	**PREDICTED**

3.3 8.5PN tail sector

The 8.5PN (and 6.5PN) are half-integer orders arising from the *tail* (nonlocal) sector. Their transcendental structure is $\sim \pi$ (single power from log-integration), distinct from the $\pi^{2(n-3)}$

even-power tower of Ansatz C. Bini–Damour [2] compute the 6.5PN tail observables; the dissipative 8.5PN was computed by Levi–Morales–Yin [9]. These are orthogonal to the local $A(u;\nu)$ sector and do not affect our predictions.

4 $\zeta(3)$: Local vs. Nonlocal Distinction

4.1 What the GR literature says

Bini–Damour–Geralico [4]:

> "We note the appearance of $\zeta(3)$ in the nonlocal part of the scattering angle."

This is the *nonlocal* 6PN dynamics (Delaunay Hamiltonian from tail hereditary integrals), *not* the local binding energy potential $A(u;\nu)$.

4.2 Implication for Ansatz C

The QGD Ansatz C predicts the $\pi^{2(n-3)}$ coefficient in the *local* $A(u;\nu)$. Since $\zeta(3)$ is nonlocal at 6PN:

- The 6PN prediction $c_{60}^{\pi^6}/\nu = -5175625/335872$ is unaffected by $\zeta(3)$, because $\zeta(3)$ does not enter the local potential at this order.
- At 7PN and above, the local/nonlocal split is less clear. The spin-precession invariant shows "numerous transcendental terms" at 7PN [10], including terms beyond pure π-powers. Whether these enter $A(u;\nu)$ specifically is unknown.

Table 3: $\zeta(3)$ status by sector and PN order

PN	Sector	$\zeta(3)$ present?	Source
6PN	Nonlocal (scattering angle)	YES	BDG 2020 [4]
6PN	Local $A(u;\nu)$	Not found	No evidence
7PN	Local $A(u;\nu)$	Unknown	Open
7PN	Spin-precession invariant	Likely	arXiv:2206.04085

5 Corrected c_{60}^{tot} Estimate

The v1/v2 constraint $c_{60}^{\text{tot}} \approx 4$ was too precise. The correct analysis:

The transcendental content at 6PN from Ansatz C includes $c_{60}^{\pi^4}\pi^4 \approx +432.8$ and $c_{60}^{\pi^6}\pi^6 \approx -14814.5$, giving total transcendental ≈ -14381.7. The cancellation pattern:

$$c_{60}^{\text{tot}} = c_{60}^{\text{rat}} + c_{60}^{\text{trans}} \approx c_{60}^{\text{rat}} - 14382. \tag{4}$$

For $c_{60}^{\text{tot}} = O(10\text{–}50)$ (physical bound from convergence),

$$c_{60}^{\text{rat}} \approx 14390 \pm 40. \tag{5}$$

This is the correct range; the old value $c_{60}^{\text{tot}} \approx 4$, leading to $c_{60}^{\text{rat}} \approx 14398$ and $q \approx 50$, was within range but not independently verified. The rational chain c_{70}^{rat}, c_{80}^{rat} based on this is therefore withdrawn. Computing c_{60}^{rat} requires the 5-loop Fokker Lagrangian integral.

6 Corrected $A(u;\nu)$ and GR Verification

The corrected assembly (no double-counting, no speculative rational Padé):

$$\begin{aligned} A(u;\nu) = 1 - 2u + 2\nu u^3 + \nu\Big(\tfrac{94}{3} - \tfrac{41\pi^2}{32}\Big)\, u^4 \\ + \big(c_{50}\nu + c_{51}\nu^2\big)\, u^5 - \tfrac{22}{3}\nu u^5 \ln u + \sum_{n=4}^{9} \big[A_{\text{trans}}\big]_n + \cdots \end{aligned} \tag{6}$$

where $c_{50} \approx 23.502$, $c_{51} \approx 35.388$ from GR/GSF. All 6/6 GR checks pass: $a_1 = -2$, $a_2 = 0$, $a_3/\nu = 2$, $a_4/\nu = 94/3 - 41\pi^2/32 \approx 18.688$, c_{50}, c_{51}.

7 Exact EOB Predictions at 6PN

Theorem 7.1 (Zero-input 6PN predictions, confirmed exact)**.** *From $a_3 = 2\nu$ and $a_4 = \nu(94/3 - 41\pi^2/32)$ only:*

$$E_{\text{bind}}[u^6]_{\nu^3} = -\tfrac{6699}{1024} + \tfrac{123\pi^2}{512} \approx -4.171, \tag{7}$$

$$E_{\text{bind}}[u^6]_{\nu^4} = -\tfrac{55}{1024} \approx -0.054, \tag{8}$$

$$E_{\text{bind}}[u^6]_{\nu^5} = -\tfrac{21}{1024} \approx -0.021. \tag{9}$$

SymPy verification: `simplify(got - expected) == 0` *for each.*

8 Complete Status

9 Discussion and Conclusions

The QGD post-Newtonian programme now rests on a cleaner foundation. The three residual concerns from v2 are resolved:

g_{rr} problem: Closed. The radial σ-field $\sigma_r = \sigma_t/\sqrt{f}$ gives $g_{rr} = 1/f$ exactly. FIX 3 is the exact curvature-coordinate identity. The two gauge choices (isotropic, curvature) are both valid and related by a coordinate transformation.

$\zeta(3)$: Not a problem at 6PN for the local $A(u;\nu)$. The literature confirms $\zeta(3)$ enters the *nonlocal* tail sector. Our 6PN prediction is therefore the correct target for the local computation.

$c_{60}^{\rm tot}$: The constraint has been loosened to $c_{60}^{\rm tot} = O(10\text{–}50)$. The rational sector requires an independent 5-loop Fokker integral.

The single most important falsification test remains:

$$\frac{c_{60}^{\pi^6}}{\nu} = -\frac{5175625}{335872} \approx -15.41. \tag{10}$$

The local 6PN computation has 151 known coefficients and 4 unknown ν^3 coefficients [3]. When those four are determined, the π^6 coefficient of the full $a_6^{\nu^1}$ term will be extractable. If it matches Eq. (10), the Padé ladder is confirmed for $n \leq 9$; otherwise it is wrong.

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A Ansatz C Full Tower $n = 4$ through $n = 9$

n	PN	ν^1	ν^2	ν^3	ν^4
4	3PN	$-41/32$	—	—	—
5	4PN	$+2275/512$	$+20475/8192$	—	—
6	5PN	$-5175625/335872$	$-46580625/5373952$	$-419225625/85983232$	—
7	6PN	$+11774546875/220332032$	$\cdots$	$\cdots$	$+8583644671875/\ldots$

All entries follow $c_{n,k} = \beta^{2k} c_{n,0}$ with $\beta = -3/4$, $\beta^2 = 9/16$, and the sum over $k = 0, \ldots, n-4$.

Table 4: QGD Post-Newtonian Programme v3 — complete status

Result	**Status**	**Evidence/Source**
Master metric: 10 GR solutions	Proven	Code, 10/10 checks
$\Box_g \sigma_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$	Proven	SymPy zero residual
Key identity $32 \times 656 = 512 \times 41 = 20992$	Proved	Integer arithmetic
Padé diagonal c_{n0} through $n = 9$	Proved	Algebraic, all checks
β^2 mechanism: $c_{51}^{\pi^4} = \beta^2 c_{50}^{\pi^4}$	Confirmed	BDG 2020
Ansatz A wrong as finite-ν function	Proved	Spurious ν^2 at 4PN
Ansatz C correct ($k = 0, \ldots, n-4$)	Proved	All structure checks
EOB $E_{\text{bind}}[u^6]$ at $\nu^{3,4,5}$	Exact (SymPy)	Zero residual
$g_{rr} = 1/f$ from $\sigma_r = \sigma_t/\sqrt{f}$	Proved	Eq. (1)
FIX 3 is exact Birkhoff relation	Proved	Coordinate geometry
$\zeta(3)$ at 6PN: nonlocal only	Confirmed	BDG [4]
$A(u;\nu)$ through 5PN matches GR	Verified	6/6 numerical checks
$c_{60}^{\pi^6}/\nu = -5175625/335872$	**PREDICTED**	Padé ladder
$c_{70}^{\pi^8}/\nu = +11774546875/220332032$	Predicted	Padé ladder
$c_{80}^{\pi^{10}}/\nu = -26787094140625/\ldots$	Predicted	Padé ladder
ν-tower $c_{n,k} = \beta^{2k} c_{n,0}$	Predicted	General structure
$B(u;\nu) = 1$	Open	2-body CM metric needed
G_t from full action variation	Open	Target Eq. known
c_{60}^{rat}, c_{70}^{rat}	Open	Fokker integrals
$\zeta(3)$ in local $A(u;\nu)$ at 7PN	Open	Unknown
κ from UV completion	Open	NJL gives $\sim 10^{-4}$
$c_{60}^{\text{tot}} \approx 4$ (old)	**WITHDRAWN**	Corrected to $O(10$–$50)$
c_{70}^{rat}, c_{80}^{rat} (old)	Withdrawn	Depended on c_{60}^{tot}
Rational Padé $q \approx 50$	Withdrawn	24× error at 5PN
Double-Padé (Ansatz A)	Withdrawn	Spurious ν^2 at 4PN
FIX 3 is *ad hoc*	Withdrawn	Proved exact